

# AIRDATA MULTIMETER CERTIFICATE OF RECALIBRATION

Customer ID: 009064 S/N: M86253  
 Customer: ARIAN TAB SERVICES, INC. City: OAK HILL State: VA  
 As-Received Model #: ADM-870 Converted to Model #: \_\_\_\_\_ Order #: R191171  
 PO #: \_\_\_\_\_ Customer Eqpt ID#: \_\_\_\_\_ Calibration Due Date: \_\_\_\_\_

This instrument has been calibrated using Calibration Standards which are traceable to NIST (National Institute of Standards and Technology). Test accuracy ratio is 4:1 for pressures and temperature. Quality Assurance Program and calibration procedures meet the requirements for ANSI/NCSL Z540-1, ISO 17025, MIL-STD 45662A and manufacturer's specifications. Calibration accuracy is certified when meters are used with properly functioning accessories only. All Uncertainties are expressed in expanded terms (twice the calculated uncertainty). This report shall not be reproduced, except in full, without the written approval of Shortridge Instruments, Inc. Results relate only to the item calibrated. For limitations on use, see Shortridge Instruments, Inc. Instruction Manual for the use of AirData Multimeters. Procedure used: Procedure for Differential Pressure, Absolute Pressure and Temperature Recalibration of AirData Multimeters SIP-CP02  
 Revision: 30 Dated: 04/04/16

Calibration Technician(s): M. Ramirez P. Bobb Calibration Date: 04/26/2019  
 Calibration Approved by: J. L. Lullman Title: Cal Mgr Date: 07/26/2019

AS-Received By: (M)  
 Date: 04/18/19 Rh: 32 %  
 Ambient Temperature: 77 °F  
 Barometric Pressure: 28.62 in Hg  
 All within spec (YES) NO NA

FINAL Test By: EB  
 Date: 04/26/19 Rh: 28 %  
 Ambient Temperature: 75 °F  
 Barometric Pressure: 28.39 in Hg  
 All within spec (YES) NO

Test By: \_\_\_\_\_  
 Date: \_\_\_\_\_ Rh: \_\_\_\_\_ %  
 Ambient Temperature: 75 °F  
 Barometric Pressure: NA in Hg  
 All within spec YES NO

## ABSOLUTE PRESSURE TEST (in Hg)

TEST METER TOLERANCE =  $\pm 2.0\% \pm .1$  in Hg AS-RECEIVED TEST WITHIN SPEC (YES) NO N/A See Notes

|                                                                         |                                                                         |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Pressure Standard: Heise #02-R S/N: 41741/42451 As-Rcvd Test 2 Test 3   | Pressure Standard: Heise #12A-R S/N: 45605/48491 As-Rcvd Test 2 Test 3  |
| Pressure Standard: Heise #04-R S/N: 41743/42453 As-Rcvd Test 2 Test 3   | Pressure Standard: Heise #14-R S/N: 43412/45043-1 As-Rcvd Test 2 Test 3 |
| Pressure Standard: Heise #06-R S/N: 41742/42452-1 As-Rcvd Test 2 Test 3 | Pressure Standard: Heise #16-R S/N: 43413/45044 As-Rcvd Test 2 Test 3   |
| Pressure Standard: Heise #08-R S/N: 42186/43328 As-Rcvd Test 2 Test 3   | Pressure Standard: Heise #18-R S/N: 44581/46845 As-Rcvd Test 2 Test 3   |
| Pressure Standard: Heise #10-R S/N: 42203/43352 As-Rcvd Test 2 Test 3   | Pressure Standard: Heise #20-R S/N: 44582/46847 As-Rcvd Test 2 Test 3   |

| Approx Set Pt | Standard | Test Meter | % Diff | Standard | Test Meter | % Diff | Standard | Test Meter | % Diff |
|---------------|----------|------------|--------|----------|------------|--------|----------|------------|--------|
| 14.0          | 14.05    | 14.0       | -0.36  | 14.29    | 14.3       | .07    |          |            |        |
| 28.4          | 28.62    | 28.5       | -0.42  | 28.39    | 28.4       | .04    |          |            |        |
| 40.0          | 40.04    | 40.0       | -0.10  | 40.27    | 40.3       | .07    |          |            |        |

## DIFFERENTIAL PRESSURE TEST (in wc)

TEST METER TOLERANCE =  $\pm 2.0\% \pm 0.001$  in wc AS-RECEIVED TEST WITHIN SPEC (YES) NO N/A See Notes

|                                                                        |                                                                         |
|------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Pressure Standard: Heise #01-L S/N: 41739/42449 As-Rcvd Test 2 Test 3  | Pressure Standard: Heise #11-L S/N: 43165/44551-1 As-Rcvd Test 2 Test 3 |
| Pressure Standard: Heise #01-R S/N: 41739/42446 As-Rcvd Test 2 Test 3  | Pressure Standard: Heise #11-R S/N: 43165/44730 As-Rcvd Test 2 Test 3   |
| Pressure Standard: Heise #02-L S/N: 41741/42454 As-Rcvd Test 2 Test 3  | Pressure Standard: Heise #12A-L S/N: 45605/48490 As-Rcvd Test 2 Test 3  |
| Pressure Standard: Heise #03A-L S/N: 45570/48461 As-Rcvd Test 2 Test 3 | Pressure Standard: Heise #13-L S/N: 43415/45041 As-Rcvd Test 2 Test 3   |
| Pressure Standard: Heise #03A-R S/N: 45570/48460 As-Rcvd Test 2 Test 3 | Pressure Standard: Heise #13-R S/N: 43415/45039 As-Rcvd Test 2 Test 3   |
| Pressure Standard: Heise #04-L S/N: 41743/42456 As-Rcvd Test 2 Test 3  | Pressure Standard: Heise #14-L S/N: 43412/45045 As-Rcvd Test 2 Test 3   |
| Pressure Standard: Heise #05-L S/N: 41740/42450 As-Rcvd Test 2 Test 3  | Pressure Standard: Heise #15-L S/N: 43416/45042 As-Rcvd Test 2 Test 3   |
| Pressure Standard: Heise #05-R S/N: 41740/42447 As-Rcvd Test 2 Test 3  | Pressure Standard: Heise #15-R S/N: 43416/45040 As-Rcvd Test 2 Test 3   |
| Pressure Standard: Heise #06-L S/N: 41742/42455 As-Rcvd Test 2 Test 3  | Pressure Standard: Heise #16-L S/N: 43413/45046 As-Rcvd Test 2 Test 3   |
| Pressure Standard: Heise #07-L S/N: 42185/42186 As-Rcvd Test 2 Test 3  | Pressure Standard: Heise #17-L S/N: 44579/46842 As-Rcvd Test 2 Test 3   |
| Pressure Standard: Heise #07-R S/N: 42185/43326 As-Rcvd Test 2 Test 3  | Pressure Standard: Heise #17-R S/N: 44579/46841 As-Rcvd Test 2 Test 3   |
| Pressure Standard: Heise #08-L S/N: 42186/43329 As-Rcvd Test 2 Test 3  | Pressure Standard: Heise #18-L S/N: 44581/46846 As-Rcvd Test 2 Test 3   |
| Pressure Standard: Heise #09-L S/N: 42202/43351 As-Rcvd Test 2 Test 3  | Pressure Standard: Heise #19-L S/N: 44580/46844 As-Rcvd Test 2 Test 3   |
| Pressure Standard: Heise #09-R S/N: 42202/43350 As-Rcvd Test 2 Test 3  | Pressure Standard: Heise #19-R S/N: 44580/46843 As-Rcvd Test 2 Test 3   |
| Pressure Standard: Heise #10-L S/N: 42203/43353 As-Rcvd Test 2 Test 3  | Pressure Standard: Heise #20-L S/N: 44582/46848 As-Rcvd Test 2 Test 3   |

| Approx Set Pt | Standard | Test Meter | % Diff | Standard | Test Meter | % Diff | Standard | Test Meter | % Diff |
|---------------|----------|------------|--------|----------|------------|--------|----------|------------|--------|
| .0500         | .0502    | .0501      | -0.20  | .0507    | .0508      | .20    |          |            |        |
| .1250         | .1256    | .1255      | -0.08  | .1257    | .1256      | -0.08  |          |            |        |
| .2250         | .2255    | .2250      | -0.22  | .2254    | .2253      | -0.04  |          |            |        |
| 1.000         | 1.047    | 1.047      | 0      | 1.015    | 1.018      | .30    |          |            |        |
| 2.000         | 2.012    | 2.016      | .20    | 2.010    | 2.013      | .15    |          |            |        |
| 3.600         | 3.653    | 3.650      | -0.08  | 3.619    | 3.619      | 0      |          |            |        |
| 4.400         | 4.410    | 4.407      | -0.07  | 4.411    | 4.415      | .09    |          |            |        |
| 27.00         | 27.33    | 27.47      | .51    | 27.19    | 27.18      | -0.04  |          |            |        |
| 50.00         | 50.74    | 50.77      | .06    | 50.02    | 49.72      | -0.60  |          |            |        |
| Overrange     | NA       | ✓          | NA     | NA       | ✓          | NA     | NA       |            | NA     |

**Shortridge Instruments, Inc.**

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